Cleaning Airbnb open dataset using Python & Pandas

In [1]:

```
# Import pandas and Load the Airbnb dataset
import pandas as pd
data=pd.read_csv('D:\Suraj\Data Analytics\Data Cleaning\Airbnb_Open_Data.csv', low_memory=False)
```

In [4]:

```
# Display first few rows
data.head()
```

Out[4]:

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	...	service fee	minimum nights
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	...	$193	10.0
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	...	$28	30.0
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	...	$124	3.0
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95976	United States	...	$74	30.0
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	...	$41	10.0

5 rows × 26 columns

In [9]:

```
# Dataset information
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 102599 entries, 0 to 102598
Data columns (total 26 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   id                              102599 non-null  int64
 1   NAME                            102349 non-null  object
 2   host id                         102599 non-null  int64
 3   host_identity_verified          102310 non-null  object
 4   host name                       102193 non-null  object
 5   neighbourhood group             102570 non-null  object
 6   neighbourhood                   102583 non-null  object
 7   lat                             102591 non-null  float64
 8   long                            102591 non-null  float64
 9   country                         102067 non-null  object
 10  country code                    102468 non-null  object
 11  instant_bookable                102494 non-null  object
 12  cancellation_policy             102523 non-null  object
 13  room type                       102599 non-null  object
 14  Construction year               102385 non-null  float64
 15  price                           102352 non-null  object
 16  service fee                     102326 non-null  object
 17  minimum nights                  102190 non-null  float64
 18  number of reviews               102416 non-null  float64
 19  last review                     86706 non-null   object
 20  reviews per month               86720 non-null   float64
 21  review rate number              102273 non-null  float64
 22  calculated host listings count  102280 non-null  float64
 23  availability 365                102151 non-null  float64
 24  house_rules                     50468 non-null   object
 25  license                         2 non-null       object
dtypes: float64(9), int64(2), object(15)
memory usage: 20.4+ MB
```

In [12]:

```
#Statstical Summery
data.describe().round()
```

Out[12]:

	id	host id	lat	long	Construction year	minimum nights	number of reviews	reviews per month	review rate number	calculated host listings count	availability 365
count	102599.0	1.025990e+05	102591.0	102591.0	102385.0	102190.0	102416.0	86720.0	102273.0	102280.0	102151.0
mean	29146235.0	4.925411e+10	41.0	-74.0	2012.0	8.0	27.0	1.0	3.0	8.0	141.0
std	16257506.0	2.853900e+10	0.0	0.0	6.0	31.0	50.0	2.0	1.0	32.0	135.0
min	1001254.0	1.236005e+08	40.0	-74.0	2003.0	-1223.0	0.0	0.0	1.0	1.0	-10.0
25%	15085814.0	2.458333e+10	41.0	-74.0	2007.0	2.0	1.0	0.0	2.0	1.0	3.0
50%	29136603.0	4.911774e+10	41.0	-74.0	2012.0	3.0	7.0	1.0	3.0	1.0	96.0
75%	43201198.0	7.399650e+10	41.0	-74.0	2017.0	5.0	30.0	2.0	4.0	2.0	269.0
max	57367417.0	9.876313e+10	41.0	-74.0	2022.0	5645.0	1024.0	90.0	5.0	332.0	3677.0

In [3]:

```
#Identify Missing Values
data.isna().sum()
```

```
long                                8
country                           532
country code                      131
instant_bookable                  105
cancellation_policy                76
room type                           0
Construction year                 214
price                             247
service fee                       273
minimum nights                    409
number of reviews                 183
last review                     15893
reviews per month               15879
review rate number                326
calculated host listings count    319
availability 365                  448
house_rules                     52131
license                        102597
dtype: int64
```

In [3]:
```
data.columns
```

Out[3]:
```
Index(['id', 'NAME', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review', 'reviews per month',
       'review rate number', 'calculated host listings count',
       'availability 365', 'house_rules', 'license'],
      dtype='object')
```

In [5]:
```
data["license"]
```

Out[5]:
```
0         NaN
1         NaN
2         NaN
3         NaN
4         NaN
         ...
102594    NaN
102595    NaN
102596    NaN
102597    NaN
102598    NaN
Name: license, Length: 102599, dtype: object
```

Task to be performed:

1.Deleting redundant columns

2.Renaming the columns

3.Dropping duplicates

4.Remove NaN values from the dataset

5.Cleaning indivisual columns

6.Check for some more Tranformations

1.Deleting redundant columns

In [6]:
```
columns_to_keep=['NAME', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review']

columns_to_drop=['reviews per month',
       'review rate number', 'calculated host listings count',
       'availability 365', 'house_rules', 'license','id']
```

In [7]:
```
len(columns_to_keep)
```

Out[7]: 19

In [8]:
```
len(columns_to_drop)
```

Out[8]: 7

1st Method, Filtering Data

In [9]:
```
data[columns_to_keep]
```

Out[9]:

	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	country code	instant_book
0	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	US	F
1	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	US	F
2	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	US	
3	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95976	United States	US	
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	US	F

In [10]:
```
data.shape
```

Out[10]: (102599, 26)

In [11]:

```
df=data[columns_to_keep]
df
```

Out[11]:

	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	country code	instant_bookab
0	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	US	Fals
1	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	US	Fals
2	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	US	Tru
3	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95976	United States	US	Tru
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	US	Fals
...	...	...	...	...	...	...	...	...	...	...	
102594	Spare room in Williamsburg	12312296767	verified	Krik	Brooklyn	Williamsburg	40.70862	-73.94651	United States	US	Fals
102595	Best Location near Columbia U	77864383453	unconfirmed	Mifan	Manhattan	Morningside Heights	40.80460	-73.96545	United States	US	Tru
102596	Comfy, bright room in Brooklyn	69050334417	unconfirmed	Megan	Brooklyn	Park Slope	40.67505	-73.98045	United States	US	Tru
102597	Big Studio-One Stop from Midtown	11160591270	unconfirmed	Christopher	Queens	Long Island City	40.74989	-73.93777	United States	US	Tru
102598	585 sf Luxury Studio	68170633372	unconfirmed	Rebecca	Manhattan	Upper West Side	40.76807	-73.98342	United States	US	Fals

102599 rows × 19 columns

2nd Method, Dropping Columns

In [12]:
```
data.columns
```

Out[12]:
```
Index(['id', 'NAME', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review', 'reviews per month',
       'review rate number', 'calculated host listings count',
       'availability 365', 'house_rules', 'license'],
      dtype='object')
```

In [13]:
```
data.drop(columns=columns_to_drop,inplace=True)
```

In [14]:
```
data.shape
```

Out[14]:
```
(102599, 19)
```

2.Renaming the columns

In [15]:
```
data.columns
```

Out[15]:
```
Index(['NAME', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review'],
      dtype='object')
```

In [16]:
```
data.rename(columns={"NAME":"Name"},inplace=True)
```

In [17]:
```
data.columns

```

Out[17]:
```
Index(['Name', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review'],
      dtype='object')
```

In [18]:
```
new_columns_names=[]
for i in data.columns:
   new_columns_names.append(i.upper())
```

In [19]:
```
new_columns_names
```

Out[19]:
```
['NAME',
 'HOST ID',
 'HOST_IDENTITY_VERIFIED',
 'HOST NAME',
 'NEIGHBOURHOOD GROUP',
 'NEIGHBOURHOOD',
 'LAT',
 'LONG',
 'COUNTRY',
 'COUNTRY CODE',
 'INSTANT_BOOKABLE',
 'CANCELLATION_POLICY',
 'ROOM TYPE',
 'CONSTRUCTION YEAR',
 'PRICE',
 'SERVICE FEE',
 'MINIMUM NIGHTS',
 'NUMBER OF REVIEWS',
 'LAST REVIEW']
```

In [20]:
```
new_dataframe_to_modify=data.copy()
new_dataframe_to_modify
```

Out[20]:

	Name	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	country code	instant_book
0	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	US	F
1	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	US	F
2	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	US	
3	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95976	United States	US	
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	US	F

In [21]:
```
new_dataframe_to_modify.columns=new_columns_names
```

In [22]:
```
new_dataframe_to_modify.columns
```

Out[22]:
```
Index(['NAME', 'HOST ID', 'HOST_IDENTITY_VERIFIED', 'HOST NAME',
       'NEIGHBOURHOOD GROUP', 'NEIGHBOURHOOD', 'LAT', 'LONG', 'COUNTRY',
       'COUNTRY CODE', 'INSTANT_BOOKABLE', 'CANCELLATION_POLICY', 'ROOM TYPE',
       'CONSTRUCTION YEAR', 'PRICE', 'SERVICE FEE', 'MINIMUM NIGHTS',
       'NUMBER OF REVIEWS', 'LAST REVIEW'],
      dtype='object')
```

In [23]:
```
data.columns
```

Out[23]:
```
Index(['Name', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review'],
      dtype='object')
```

3.Dropping duplicates

In [24]:
```
data.duplicated().sum()
```

Out[24]:
```
541
```

In [25]:
```
data.duplicated().value_counts()
```

Out[25]:
```
False    102058
True        541
dtype: int64
```

In [11]:

```
duplicates=data[data.duplicated(keep=False)].sort_values(by=list(data.columns))
duplicates
```

Out[11]:

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	...	service fee	minim nig
9098	6026161	Upper East Side 2 bedroom- close to Hospitals-	65193709566	verified	Juliana	Manhattan	Upper East Side	40.76222	-73.96030	United States	...	$21	3
102474	6026161	Upper East Side 2 bedroom- close to Hospitals-	65193709566	verified	Juliana	Manhattan	Upper East Side	40.76222	-73.96030	United States	...	$21	3
9099	6026714	Close to East Side Hospitals- Modern 2 Bedroom...	31072202372	verified	Juliana	Manhattan	Upper East Side	40.76249	-73.96217	United States	...	$57	3
102475	6026714	Close to East Side Hospitals- Modern 2 Bedroom...	31072202372	verified	Juliana	Manhattan	Upper East Side	40.76249	-73.96217	United States	...	$57	3
9100	6027266	ACADIA Spacious 2 Bedroom Apt - Close to Hospi...	95854111798	verified	Juliana	Manhattan	Upper East Side	40.76021	-73.96157	United States	...	$117	3
...	...	...	...	...	...	...	...	...	...	...	...	...	
102239	35606797	Bright and Beautiful Top Floor Two Bedrooms	65331079885	unconfirmed	Frankie	Brooklyn	Carroll Gardens	40.68383	-73.99281	United States	...	$205	3
62658	35607349	Modern & Bright Queen Bedroom Midtown East	57770451783	unconfirmed	David	Manhattan	Upper East Side	40.76132	-73.96064	United States	...	$28	3
102240	35607349	Modern & Bright Queen Bedroom Midtown East	57770451783	unconfirmed	David	Manhattan	Upper East Side	40.76132	-73.96064	United States	...	$28	3
62659	35607902	Modern NEW Room|PRIVATE BATHROOM	7431680152	verified	Dariia & Jacob	Brooklyn	Bedford-Stuyvesant	40.68990	-73.94074	United States	...	$57	3

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	...	service fee	minim nig
102241	35607902	Modern NEW Room|PRIVATE BATHROOM	7431680152	verified	Dariia & Jacob	Brooklyn	Bedford-Stuyvesant	40.68990	-73.94074	United States	...	$57	3

1082 rows × 26 columns

In [27]:
```
data.drop_duplicates(inplace=True)
data.duplicated().sum()
```

Out[27]:
```
0
```

In [28]:
```
data.isna().sum()
```

Out[28]:
```
Name                      250
host id                     0
host_identity_verified    289
host name                 404
neighbourhood group        29
neighbourhood              16
lat                         8
long                        8
country                   532
country code              131
instant_bookable          105
cancellation_policy        76
room type                   0
Construction year         214
price                     247
service fee               273
minimum nights            400
number of reviews         183
last review             15832
dtype: int64
```

4.Remove NaN values from the dataset

In [29]:
```
data['last review'].isna().sum()
```

Out[29]:
```
15832
```

In [30]:
```
data.drop(columns=["last review"],inplace=True)
```

In [32]:
```
data.isna().sum()
```

Out[32]:
```
Name                      250
host id                     0
host_identity_verified    289
host name                 404
neighbourhood group        29
neighbourhood              16
lat                         8
long                        8
country                   532
country code              131
instant_bookable          105
cancellation_policy        76
room type                   0
Construction year         214
price                     247
service fee               273
minimum nights            400
number of reviews         183
dtype: int64
```

In [33]:
```
data.dropna(inplace=True)
```

In [34]:
```
data.isna().sum()
```

Out[34]:
```
Name                      0
host id                   0
host_identity_verified    0
host name                 0
neighbourhood group       0
neighbourhood             0
lat                       0
long                      0
country                   0
country code              0
instant_bookable          0
cancellation_policy       0
room type                 0
Construction year         0
price                     0
service fee               0
minimum nights            0
number of reviews         0
dtype: int64
```

5.Cleaning indivisual columns

In [38]:

```
data['host_identity_verified']=data['host_identity_verified'].str.upper()
data
```

..	...	...	...	...	...	...	...	...	...	...	...	...	...
a h d s	5155155913	UNCONFIRMED	Cristina	Queens	Astoria	40.76987	-73.92679	United States	US	False	flexible	Private room	2
st g it	71564706996	VERIFIED	Petra	Brooklyn	Bedford-Stuyvesant	40.68335	-73.91222	United States	US	True	moderate	Private room	2
n g	83101851929	VERIFIED	Russell	Brooklyn	Williamsburg	40.70870	-73.96710	United States	US	True	flexible	Private room	2
s d	69037598918	UNCONFIRMED	Christine	Queens	Bayside	40.75047	-73.75349	United States	US	False	moderate	Private room	2
n ', (	56457739998	VERIFIED	Sonia	Queens	East Elmhurst	40.76245	-73.87938	United States	US	True	strict	Private room	2

mns

In [42]:

```
data['instant_bookable']=data['instant_bookable'].apply(lambda x:1 if x==True else 0)
data
```

Out[42]:

	Name	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	country code	instant_bookable
0	Clean & quiet apt home by the park	80014485718	UNCONFIRMED	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	US	0
1	Skylit Midtown Castle	52335172823	VERIFIED	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	US	0
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	VERIFIED	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	US	0
5	Large Cozy 1 BR Apartment In Midtown East	45498551794	VERIFIED	Michelle	Manhattan	Murray Hill	40.74767	-73.97500	United States	US	1
7	BlissArtsSpace!	90821839709	UNCONFIRMED	Emma	Brooklyn	Bedford-Stuyvesant	40.68688	-73.95596	United States	US	0
...	...	...	...	...	...	...	...	...	...	...	...
102029	Cozy room in a 2B with backyard access	5155155913	UNCONFIRMED	Cristina	Queens	Astoria	40.76987	-73.92679	United States	US	0
102030	An East Williamsburg Delight	71564706996	VERIFIED	Petra	Brooklyn	Bedford-Stuyvesant	40.68335	-73.91222	United States	US	1
102031	Cozy bedroom in williamsburg	83101851929	VERIFIED	Russell	Brooklyn	Williamsburg	40.70870	-73.96710	United States	US	1
102032	Spacious Room w/ 2 Bed	69037598918	UNCONFIRMED	Christine	Queens	Bayside	40.75047	-73.75349	United States	US	0
102040	Room in Queens, NY, near LGA.	56457739998	VERIFIED	Sonia	Queens	East Elmhurst	40.76245	-73.87938	United States	US	1

99342 rows × 18 columns

In [49]:

```
type (data["instant_bookable"][5])
```

Out[49]:

```
numpy.int64
```

In [44]:

```
data.reset_index(inplace=True)
data
```

Out[44]:

	level_0	index	Name	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	country code	ins
0	0	0	Clean & quiet apt home by the park	80014485718	UNCONFIRMED	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	US	
1	1	1	Skylit Midtown Castle	52335172823	VERIFIED	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	US	
2	2	4	Entire Apt: Spacious Studio/Loft by central park	92037596077	VERIFIED	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	US	
3	3	5	Large Cozy 1 BR Apartment In Midtown East	45498551794	VERIFIED	Michelle	Manhattan	Murray Hill	40.74767	-73.97500	United States	US	
4	4	7	BlissArtsSpace!	90821839709	UNCONFIRMED	Emma	Brooklyn	Bedford-Stuyvesant	40.68688	-73.95596	United States	US	
...	...	...	...	...	...	...	...	...	...	...	...	...	
99337	99337	102029	Cozy room in a 2B with backyard access	5155155913	UNCONFIRMED	Cristina	Queens	Astoria	40.76987	-73.92679	United States	US	
99338	99338	102030	An East Williamsburg Delight	71564706996	VERIFIED	Petra	Brooklyn	Bedford-Stuyvesant	40.68335	-73.91222	United States	US	
99339	99339	102031	Cozy bedroom in williamsburg	83101851929	VERIFIED	Russell	Brooklyn	Williamsburg	40.70870	-73.96710	United States	US	
99340	99340	102032	Spacious Room w/ 2 Bed	69037598918	UNCONFIRMED	Christine	Queens	Bayside	40.75047	-73.75349	United States	US	
99341	99341	102040	Room in Queens, NY, near LGA.	56457739998	VERIFIED	Sonia	Queens	East Elmhurst	40.76245	-73.87938	United States	US	

99342 rows × 20 columns

In [52]:
```
data.drop(columns=["index","level_0"],inplace=True)
```

...

In [53]:
```
data
```

						Stuyvesant			States		
...	...	...	...	...	...	...	...	...	...	...	...
99337	Cozy room in a 2B with backyard access	5155155913	UNCONFIRMED	Cristina	Queens	Astoria	40.76987	-73.92679	United States	US	0
99338	An East Williamsburg Delight	71564706996	VERIFIED	Petra	Brooklyn	Bedford-Stuyvesant	40.68335	-73.91222	United States	US	1
99339	Cozy bedroom in williamsburg	83101851929	VERIFIED	Russell	Brooklyn	Williamsburg	40.70870	-73.96710	United States	US	1
99340	Spacious Room w/ 2 Bed	69037598918	UNCONFIRMED	Christine	Queens	Bayside	40.75047	-73.75349	United States	US	0
99341	Room in Queens, NY, near LGA.	56457739998	VERIFIED	Sonia	Queens	East Elmhurst	40.76245	-73.87938	United States	US	1

99342 rows x 18 columns

6.Check for some more Tranformations

In [57]:
```
data['price'][15]

```

Out[57]: '$1,020 '

In [58]:
```
type(data['price'][15])
```

Out[58]: str

In [66]:
```
data['price'].head(5)
```

Out[66]:
```
0      966
1      142
2      204
3      577
4    1,060
Name: price, dtype: object
```

In [63]:
```
data['price']=data['price'].str.replace('$',"")
```

```
C:\Users\ADMIN\AppData\Local\Temp\ipykernel_9312\1951798093.py:1: FutureWarning: The default value of regex will change from Tr
ue to False in a future version. In addition, single character regular expressions will *not* be treated as literal strings whe
n regex=True.
  data['price']=data['price'].str.replace('$',"")
```

In [64]:
```
data['price'].head(3)
```

Out[64]:
```
0    966
1    142
2    204
Name: price, dtype: object
```

In [67]:
```
data['price']=data['price'].str.replace(',',"")
```

In [69]:
```
data['price'][15]
```

Out[69]:
```
'1020 '
```

In [71]:
```
data['price']=data['price'].str.replace(' ',"")
```

In [72]:
```
data['price'][15]
```

Out[72]:
```
'1020'
```

In [75]:
```
data['price']=data['price'].astype(int)
type(data['price'][15])
```

Out[75]:
```
numpy.int32
```

In [83]:

```
#save cleaned file
data.to_csv("Cleaned_Airbnb_Open_Data.csv",index=False)
```

In [84]:

```
pd.read_csv("Cleaned_Airbnb_Open_Data.csv")
```

0	Clean & quiet apt home by the park	80014485718	UNCONFIRMED	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	US	0
1	Skylit Midtown Castle	52335172823	VERIFIED	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	US	0
2	Entire Apt: Spacious Studio/Loft by central park	92037596077	VERIFIED	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	US	0
3	Large Cozy 1 BR Apartment In Midtown East	45498551794	VERIFIED	Michelle	Manhattan	Murray Hill	40.74767	-73.97500	United States	US	1
4	BlissArtsSpace!	90821839709	UNCONFIRMED	Emma	Brooklyn	Bedford-Stuyvesant	40.68688	-73.95596	United States	US	0
...	...	...	...	...	...	...	...	...	...	...	...

In [87]:

```
data.to_excel("Cleaned_Airbnb_Open_Data_excel.xlsx",index=False)
```

In [88]:

```
pd.read_excel("Cleaned_Airbnb_Open_Data_excel.xlsx").head(5)
```

Out[88]:

	Name	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	country code	instant_bookable	cance
0	Clean & quiet apt home by the park	80014485718	UNCONFIRMED	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	US	0	
1	Skylit Midtown Castle	52335172823	VERIFIED	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	US	0	
2	Entire Apt: Spacious Studio/Loft by central park	92037596077	VERIFIED	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	US	0	
3	Large Cozy 1 BR Apartment In Midtown East	45498551794	VERIFIED	Michelle	Manhattan	Murray Hill	40.74767	-73.97500	United States	US	1	
4	BlissArtsSpace!	90821839709	UNCONFIRMED	Emma	Brooklyn	Bedford-Stuyvesant	40.68688	-73.95596	United States	US	0	